



Fondamenti di Analisi dei Dati

from **data analysis** to **predictive techniques**

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Clustering

Understanding the underlying structure of complex datasets through intelligent grouping

Clustering

When dealing with complex datasets of multiple observations and variables, clustering helps us understand the underlying structure by grouping data points with similar characteristics into distinct clusters.



Customer Segmentation

Grouping customers based on purchasing behavior, preferences, or demographics for targeted marketing strategies.



Image Segmentation

Dividing images into meaningful segments based on similarity, aiding in object recognition and computer vision tasks.



Text Document Clustering

Grouping similar documents together for topic modeling, document organization, and information retrieval.



Audio Processing

Clustering audio data for speaker identification, music genre classification, and speech analysis.



E-commerce

Grouping products or users based on purchasing patterns to optimize recommendations and inventory.

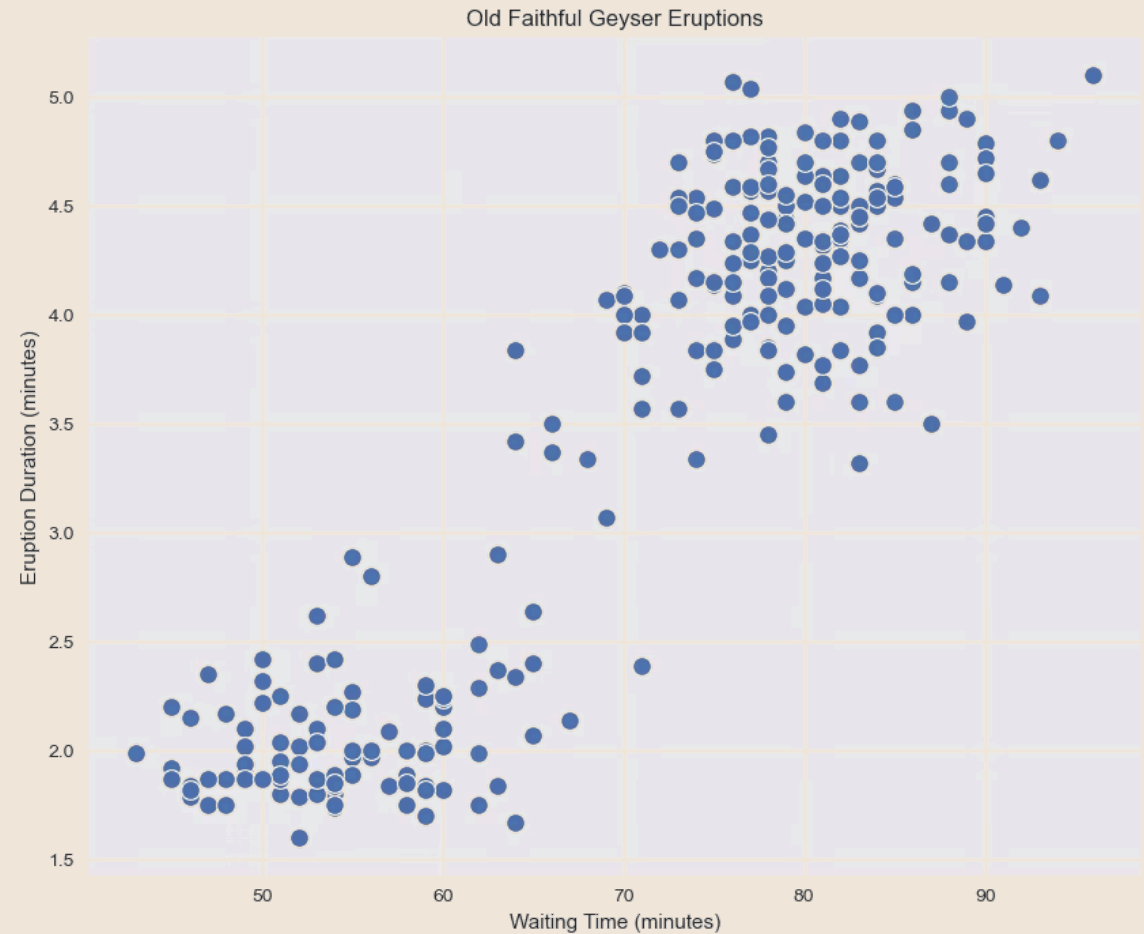
The Old Faithful Dataset

The Old Faithful geyser at Yellowstone National Park provides a perfect example of natural clustering. Each observation includes two variables:

- **Duration of eruption in minutes**
- **Time in minutes to next eruption**

When plotted, we can clearly see two distinct groups:

- short eruptions followed by quick intervals (bottom left)
- and long eruptions followed by long intervals (upper right).



Formal Problem Definition

Let

$$\mathbf{X} = \left\{ \mathbf{x}^{(i)} \right\}_{i=1}^N, \mathbf{x}^{(i)} \in \mathfrak{R}^n$$

be a set of N observations, where each observation is an n -dimensional vector. The goal of clustering is to split X into K groups (clusters)

$$S = \{S_1, S_2, \dots, S_K\}$$

Such that:

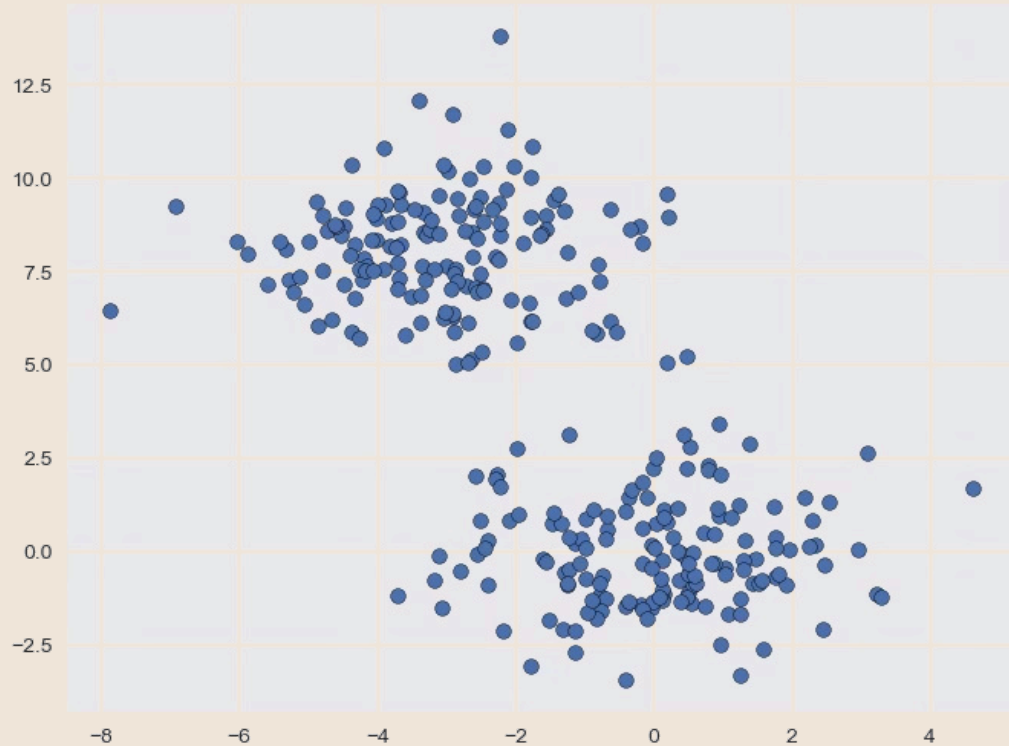
$$\forall \mathbf{x}^{(i)} \in \mathbf{X} \exists! S_j \in S : \mathbf{x}^{(i)} \in S_j$$

Every data point belongs to exactly one cluster, and elements within a group are similar to each other, while elements from different groups are dissimilar.

 **Key Insight:** The number of clusters K is often a hyper-parameter that must be specified before running the algorithm.

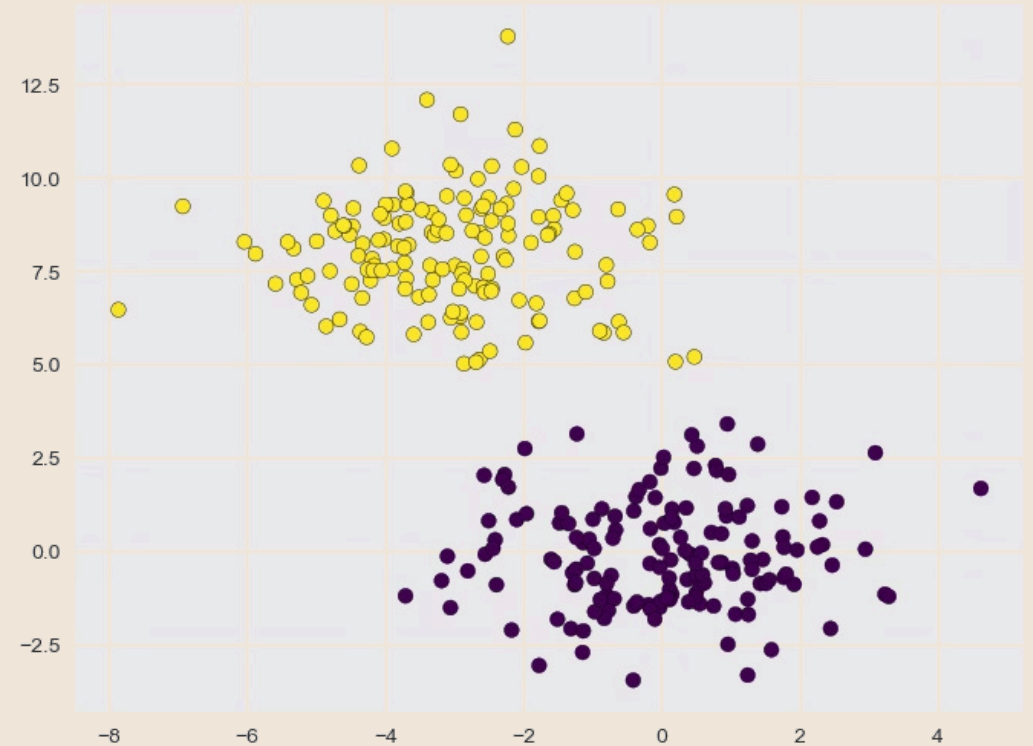
From Unlabeled to Labeled Data

Before Clustering



Data points exist without explicit group assignments. No discrete variable indicates which group each point belongs to.

After Clustering



The algorithm has identified two distinct groups, ensuring points within each group are similar while points from different groups are dissimilar.

K-Means Clustering

K-Means attempts to create K clusters which are **as compact as possible**, where K is a parameter specified by the user (its application may depend on the application).

We will start by defining K different vectors $\mu_k \in \mathbb{R}^n$ which will act as prototypes for our clusters. We will shortly see that these vectors will be the cluster's centers.

Hence, we define $N \times K$ binary variables r_{ij} which will allow us to establish a mapping between data points $\mathbf{x}^{(i)}$ and clusters S_j . In particular, we will define:

$$r_{ij} = \begin{cases} 1 & \text{if } \mathbf{x}^{(i)} \in S_j \\ 0 & \text{otherwise} \end{cases}$$

The notion that **clusters should be as compact as possible** is formalized by defining the following **cost function**, which is sometimes called also **distortion function**:

$$J = \sum_{j=1}^K \sum_{i=1}^N r_{ij} \|\mathbf{x}_i - \mu_j\|^2$$

Note that this function J is the total **Within-Cluster Sum of Squares (WCSS)**.

For a single cluster S_j , the term $\sum_{i=1}^N r_{ij} \|\mathbf{x}_i - \mu_j\|^2$ is the sum of squared distances of all points in that cluster from its center. This is a measure of the cluster's **inertia** or "compactness."

The goal of K-Means is to find the set of centroids $\{\mu_j\}$ and assignments $\{r_{ij}\}$ that minimizes this total inertia J .

The K-Means problem is solved by finding a partition $\hat{S}$ minimizing the cost function:

$$\{\hat{r}_{ij}\}_{ij}, \{\hat{\mu}_j\}_j = \arg \min \sum_{j=1}^K \sum_{i=1}^N r_{ij} \left\| \mathbf{x}^{(i)} - \mu_j \right\|^2$$

The K-Means Algorithm

1

Initialize

Randomly choose K cluster centroids $\mu_1, \mu_1, \dots, \mu_k$

2

Assignment Step

Assign each point to the nearest centroid based on Euclidean distance

3

Update Step

Recalculate centroids as the mean of all points in each cluster

4

Repeat

Iterate until convergence or maximum iterations reached

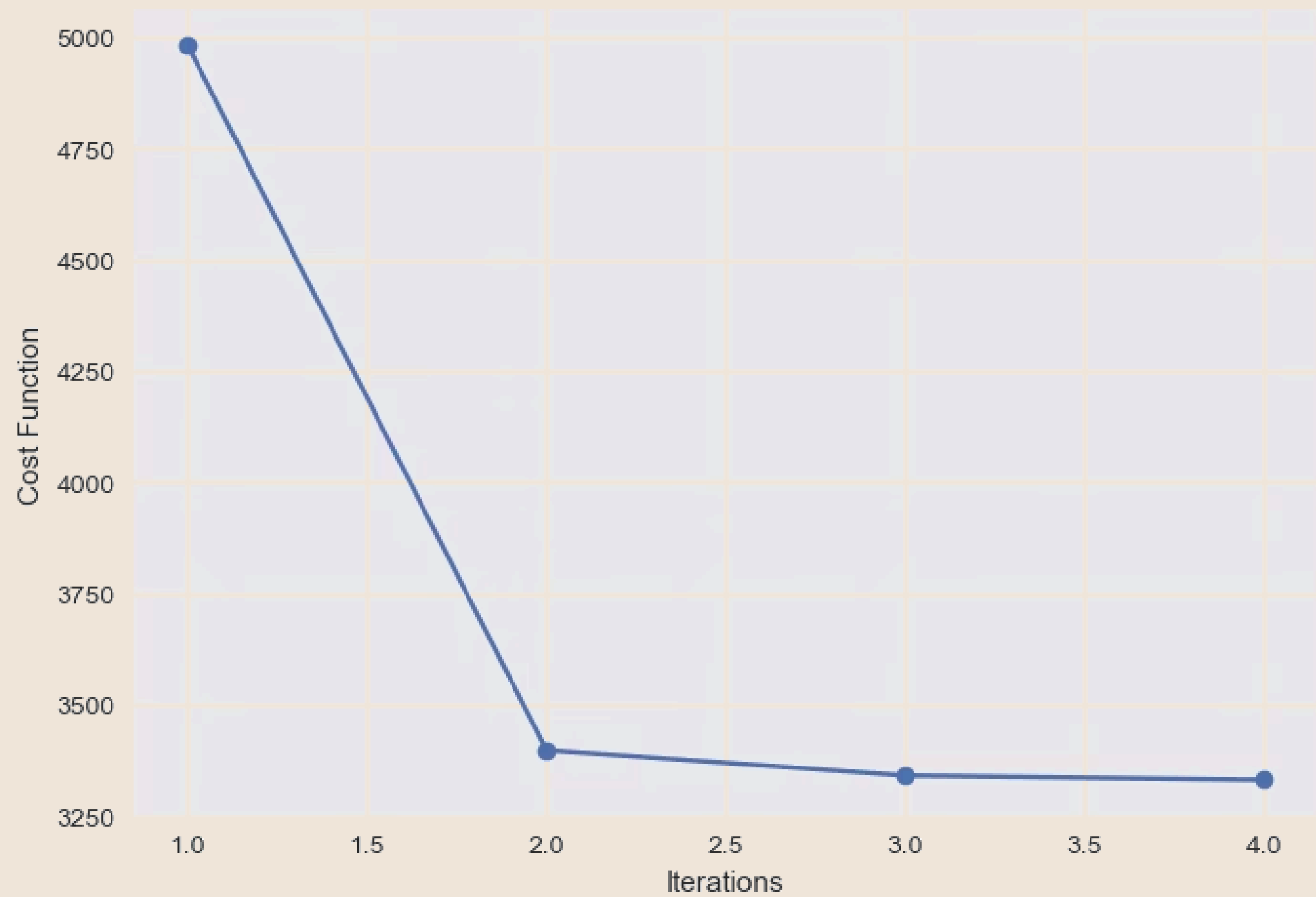
📌 **Important:** The algorithm is not guaranteed to find the global optimum—the solution may depend on random initialization. However, it usually leads to reasonable results in practice.

K-Means in Action



Convergence and Cost Function

After iteration 4, the centroids no longer move during the update step—the algorithm has converged. The data is now optimally clustered into two distinct groups with minimal within-cluster variance.



Pseudocode

Randomly initialize K cluster centroids $\mu_1, \mu_2, \dots, \mu_K \in \mathcal{R}^n$

Repeat until termination criterion is reached {

for $i = 1$ to N

$$r_{ij} = \begin{cases} 1 & \text{if } j = \arg_k \min ||\mathbf{x}^{(i)} - \mu_j||^2, \forall i, j // \text{assignment} \\ 0 & \text{otherwise} \end{cases}$$

for $j = 1$ to K

$$\mu_j = \frac{\sum_{i=1}^N r_{ij} \mathbf{x}^{(i)}}{\sum_{i=1}^N r_{ij}} // \text{update}$$

}

Choosing the Right K

The number of clusters K is a hyper-parameter that must be determined before running K-Means. Two main techniques help identify the optimal value:

Elbow Method

Plot the cost function for different K values and look for the "**elbow point**" where the rate of decrease dramatically slows. This indicates diminishing returns from adding more clusters.

Silhouette Method

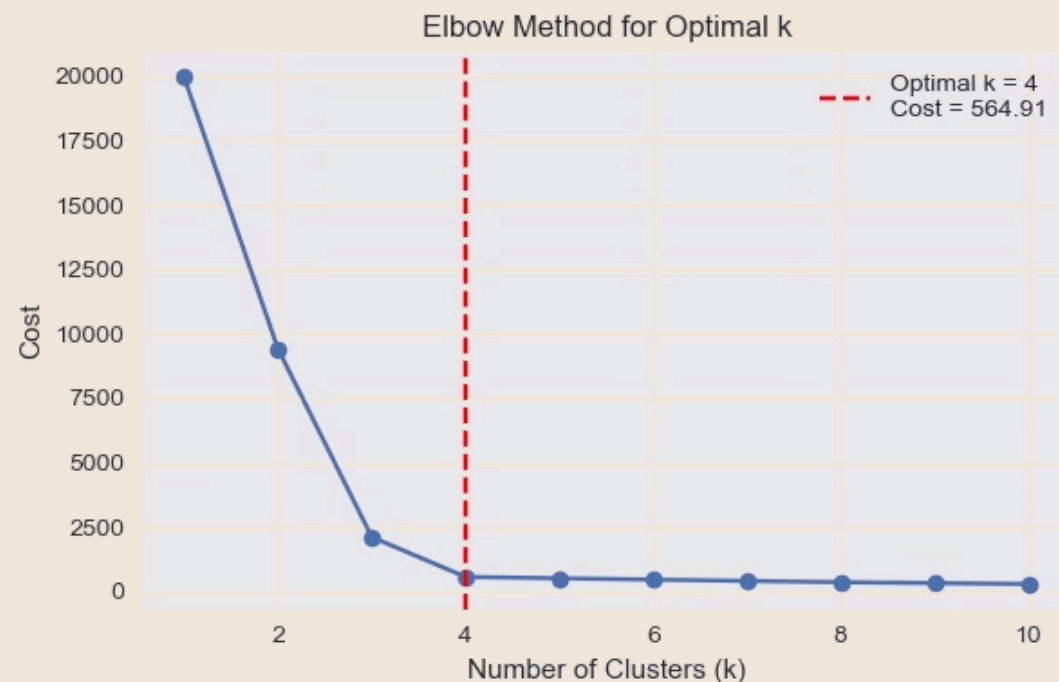
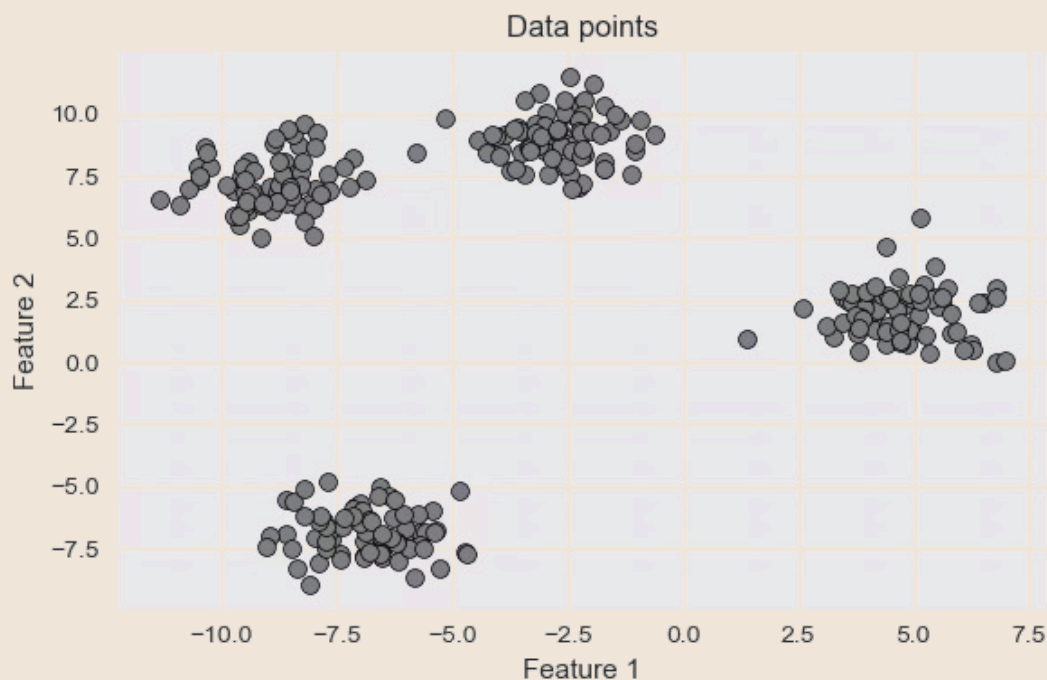
Measure **how well each point fits its assigned cluster** compared to other clusters. Higher average silhouette scores indicate better-defined, more separated clusters.

These heuristic methods don't guarantee optimal K selection, but they provide valuable guidance when combined with domain intuition.

The Elbow Method Explained

The elbow method involves fitting K-Means for different K values and plotting the final cost function. As K increases, cost decreases—more clusters naturally reduce individual cluster variance.

However, the optimal K shows a distinctive "elbow" where the cost reduction slows significantly. Beyond this point, additional clusters provide minimal improvement.



The Silhouette Method Explained

The silhouette method tries to assess how good a given clustering model is by measuring **how well the data is grouped in the identified clusters**. This is done by computing, **for each data point** a score indicating whether **how well the data point fits the assigned cluster with respect to all others**.

If the model is a good one (hence a good K value), we would expect each data point to **fit well its assigned cluster and not to fit well any other cluster**. If this does not happen, it means that the clusters are not very well separated and we **either have more or less clusters**.

To do so, for a given data point i assigned to cluster C_I , the silhouette method computes the following score:

$$a(i) = \frac{1}{|C_I| - 1} \sum_{j \in C_I, i \neq j} d(i, j)$$

where $d(i, j)$ is the Euclidean distance between data point i and data point j . In practice, the score $a(i)$ is the **average distance of data point i with any other data point assigned to the same cluster**. We expect this number to be small if the cluster has a low variance.

We now want to compare this number to the value we would have **if i were to be assigned to another cluster C_J** . In particular, we will choose the cluster C_J which minimizes the average distance. This is done by computing the following score:

$$b(i) = \min_{J \neq I} \frac{1}{|C_J|} \sum_{j \in C_J} d(i, j)$$

Now we compare $a(i)$ with $b(i)$. Ideally, $a(i)$ should be much smaller than $b(i)$. If this happens, this means that the data point i fits in the cluster C_I much better than it would in any other cluster. To do so, we compute the **silhouette score** of data point i as follows:

$$s(i) = \begin{cases} \frac{b(i) - a(i)}{\max\{a(i), b(i)\}} & \text{if } |C_I| > 1 \\ 0 & \text{otherwise} \end{cases}$$

The obtained score is such that:

$$-1 \leq s(i) \leq 1$$

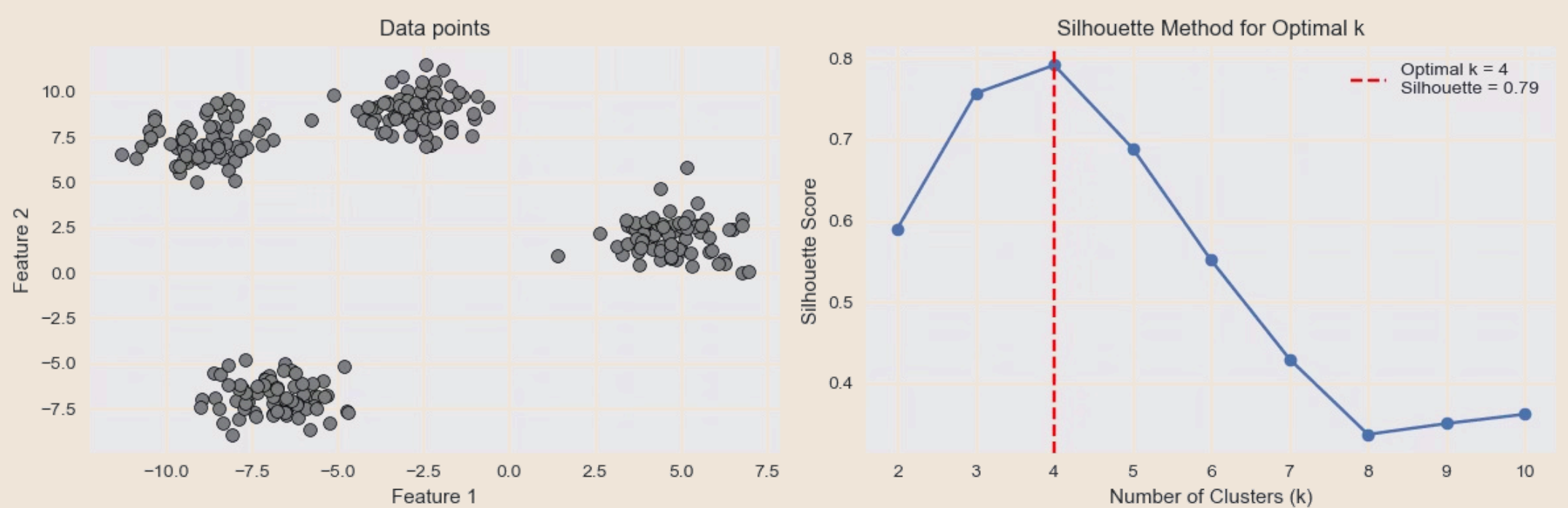
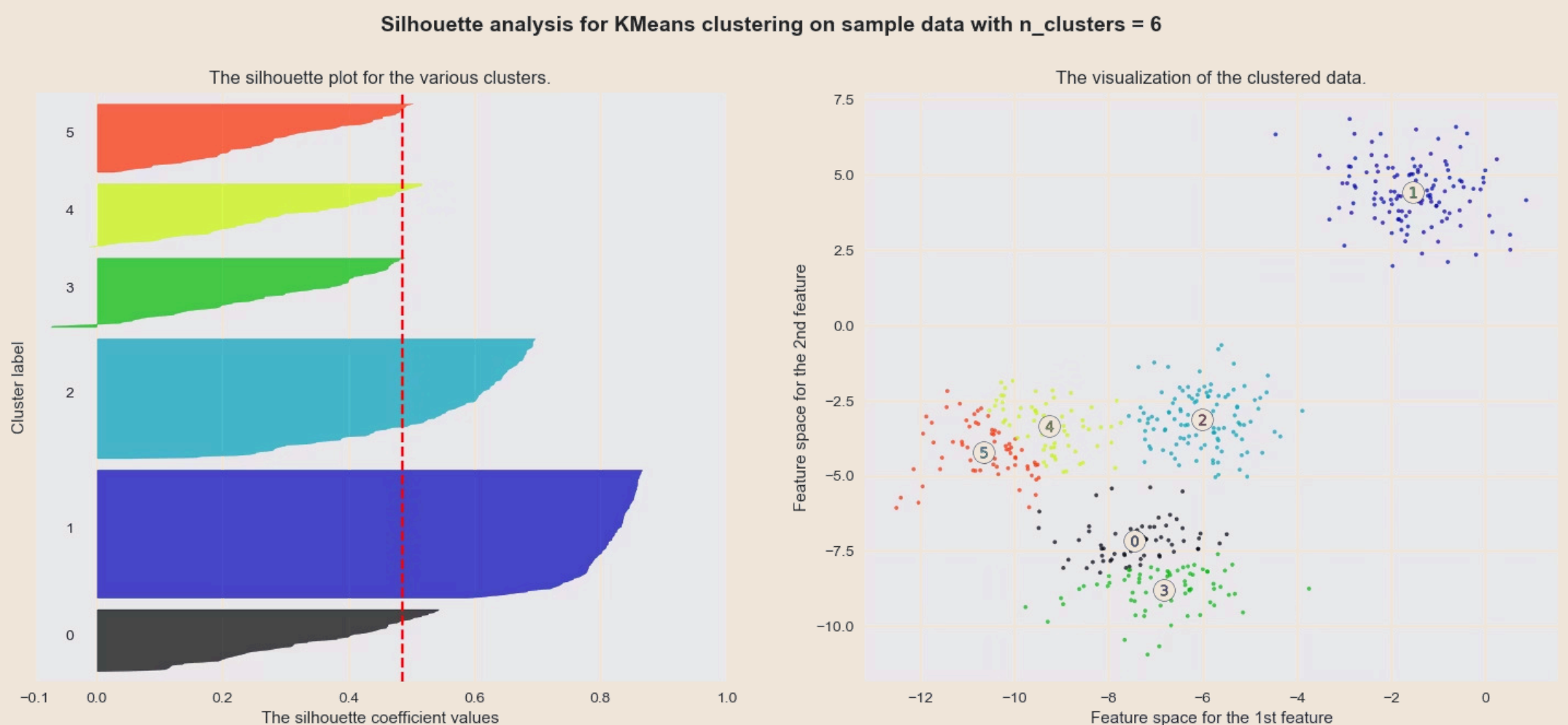
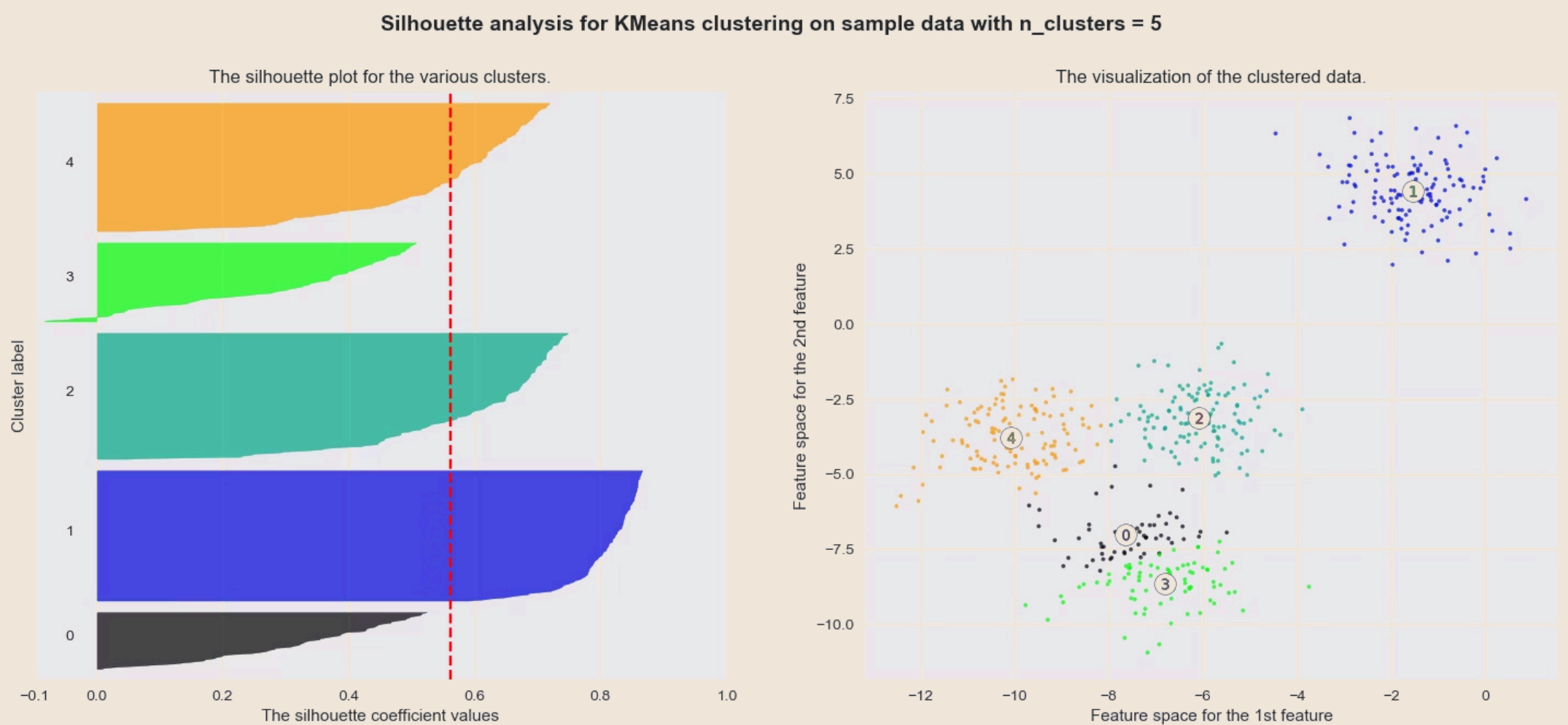
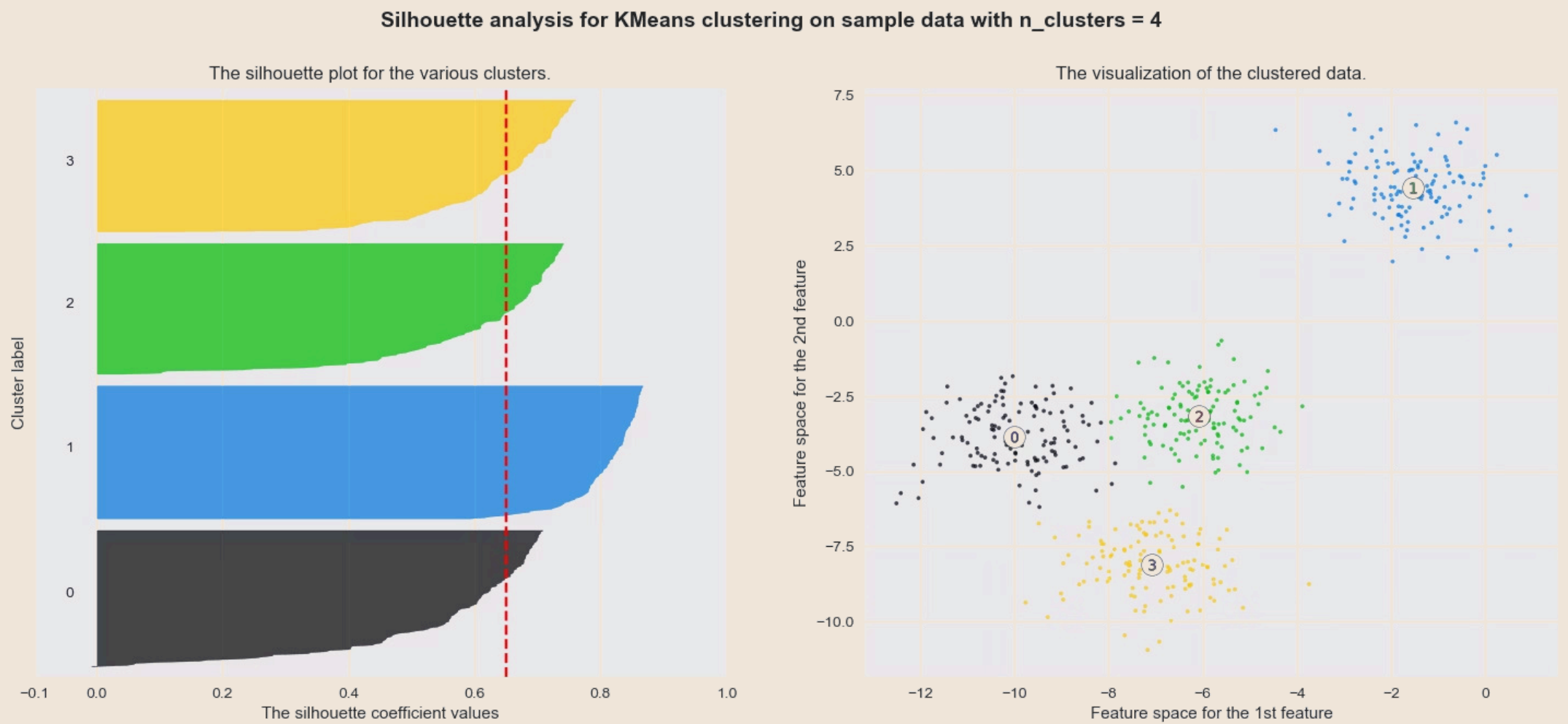
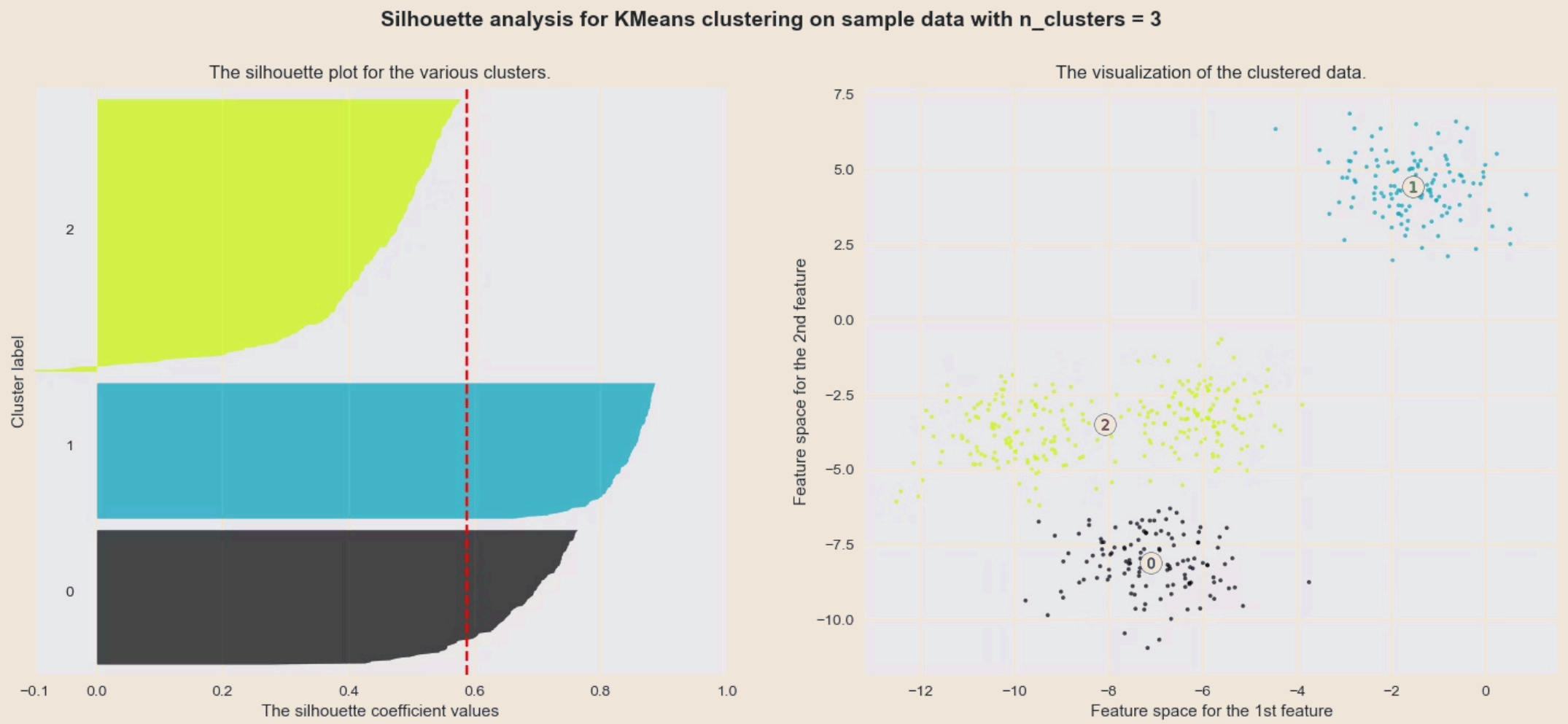
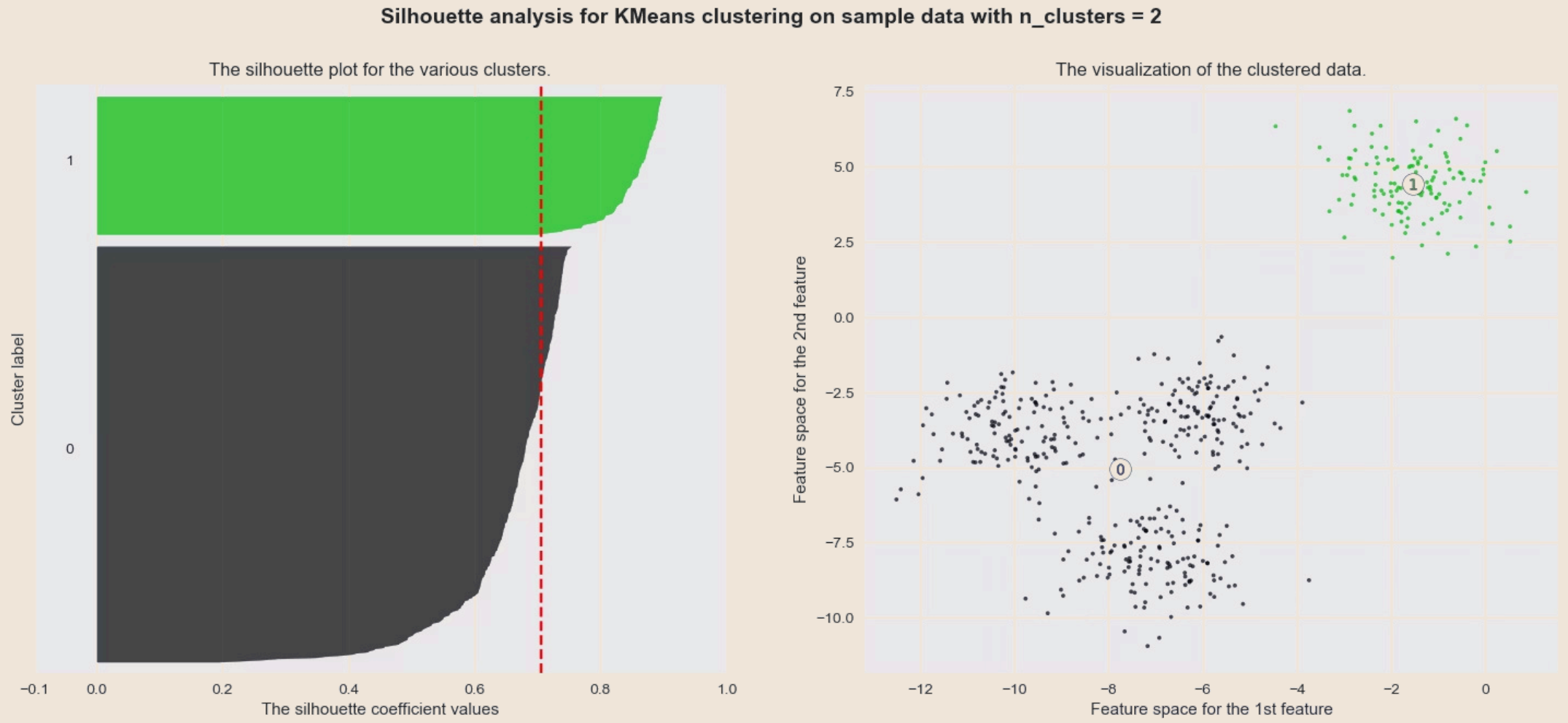
In practice:

- A value of $s(i)$ close to 1 is obtained when $a(i) \ll b(i)$, meaning that i fits in C_I much better than it would fit in any other cluster;
- A value of $s(i)$ close to -1 is obtained when $a(i) \gg b(i)$, meaning that another cluster C_J exists in which i fits better.
- A value of $s(i)$ close to 0 indicates that the data point is close to the **border between two natural clusters**.

The score above is arbitrarily set to 0 if $|C_I| = 1$ as, in that case, the cluster would only contain the data point i . Note that this is **an assumption of neutrality**.

The correct number of clusters is usually chosen by showing the silhouette plot, which displays all the silhouette scores of data points within each clusters, arranged from largest to smallest.

Silhouette Score in Action



Lab: Customer Segmentation

Let's apply K-Means to a real business problem: segmenting mall customers based on their Annual Income and Spending Score to create targeted marketing campaigns.

01

Load & Explore

Import the Mall Customers dataset and visualize relationships using pairplots.

02

Prepare & Scale

Select features and apply StandardScaler to ensure equal feature importance.

03

Find Optimal K

Use Elbow and Silhouette methods to determine the best number of clusters.

04

Fit & Interpret

Train the final model and analyze clusters from a business perspective.



Conclusions and Next Steps



We Have Explored:

- Clustering
- K-Means
- Elbow Method
- Silhouette Method

📖 In the next lectures, we will look at density estimation

References

- Section 10.3.1 of [1]
- Section 2.3.9 of [2]
- Section 9.2 of [2]
- [https://en.wikipedia.org/wiki/Silhouette_\(clustering\)](https://en.wikipedia.org/wiki/Silhouette_(clustering))

[1] James, G., Witten, D., Hastie, T., & Tibshirani, R. (2013). An introduction to statistical learning. New York: springer.

[2] Bishop, Christopher M., and Nasser M. Nasrabadi. Pattern recognition and machine learning. Vol. 4. No. 4. New York: springer, 2006.